



Module 0

Python Programming

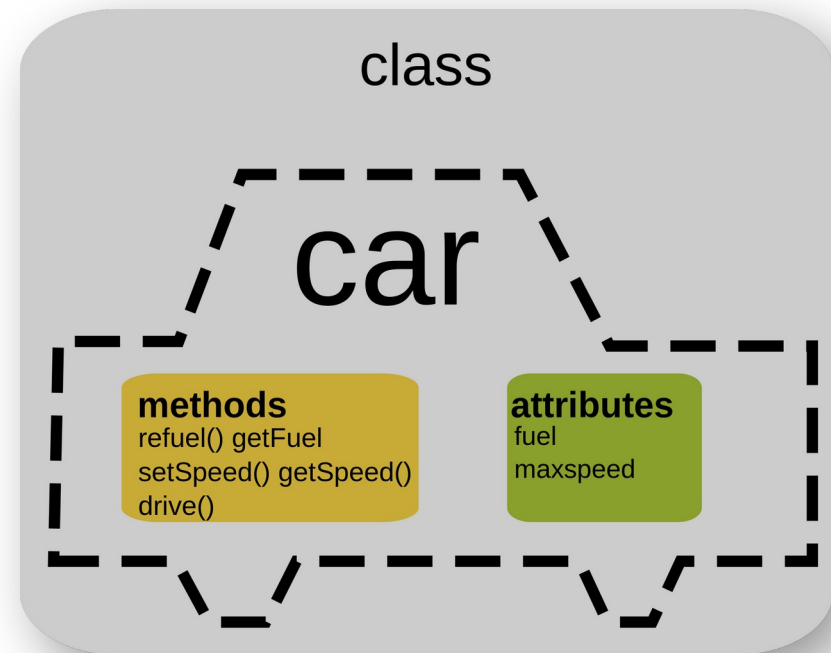


Python Basics



Objects and Classes

- ❑ Objects are encapsulations of variables and functions into single entity
- ❑ Classes provide a way to group information into a single unit
 - Can combine multiple attributes into a single object
- ❑ Classes contain methods (**functions**) and attributes (**values**)



Example

❑ Class Car

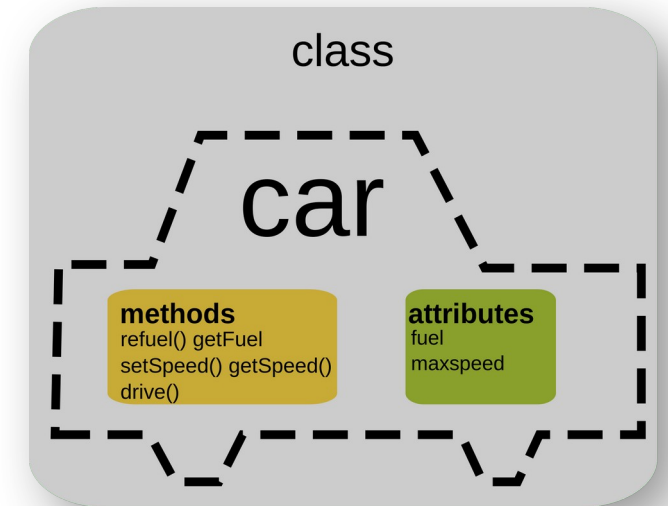


```
class Car:
    wheels = "four"
    def __init__(self, model, color):
        self.model = model
        self.color = color

    def brake(self):
        return print(f"The car is breaking...")

    def accelerate(self, speed):
        return print(f"The car is accelerating to {speed}MPH")

    def information(self):
        print(f"This is a {self.model} in color {self.color}")
```



Attributes

- ❑ Characteristics that can be accessed from other objects
- ❑ Can be mutable or immutable



```
class Person:
    fingers = 10
    def __init__(self, name, age):
        self.name = name
        self.age = age
```

Making an Instance (Instantiate)

- ❑ Creating an instance object from the class



```
class Person:  
    fingers = 10  
    def __init__(self, name, age):  
        self.name = name  
        self.age = age
```



```
p1 = Person("John", 36)
```



```
p1.age  
p1.name
```

Methods

- ❑ Enable classes to implement functions



```
import numpy as np

class stats:
    def __init__(self, list):
        self.list = list
    def compute_mean(self):
        print(np.mean(self.list))
    def compute_var(self):
        print(np.var(self.list))
```



```
my_stats = stats([1,2,3,4,5])
```

Methods

- ❑ Calling a method bound to the class



```
my_stats = stats([1,2,3,4,5])
```



```
my_stats.compute_mean()
```

3.0



```
my_stats.compute_var()
```

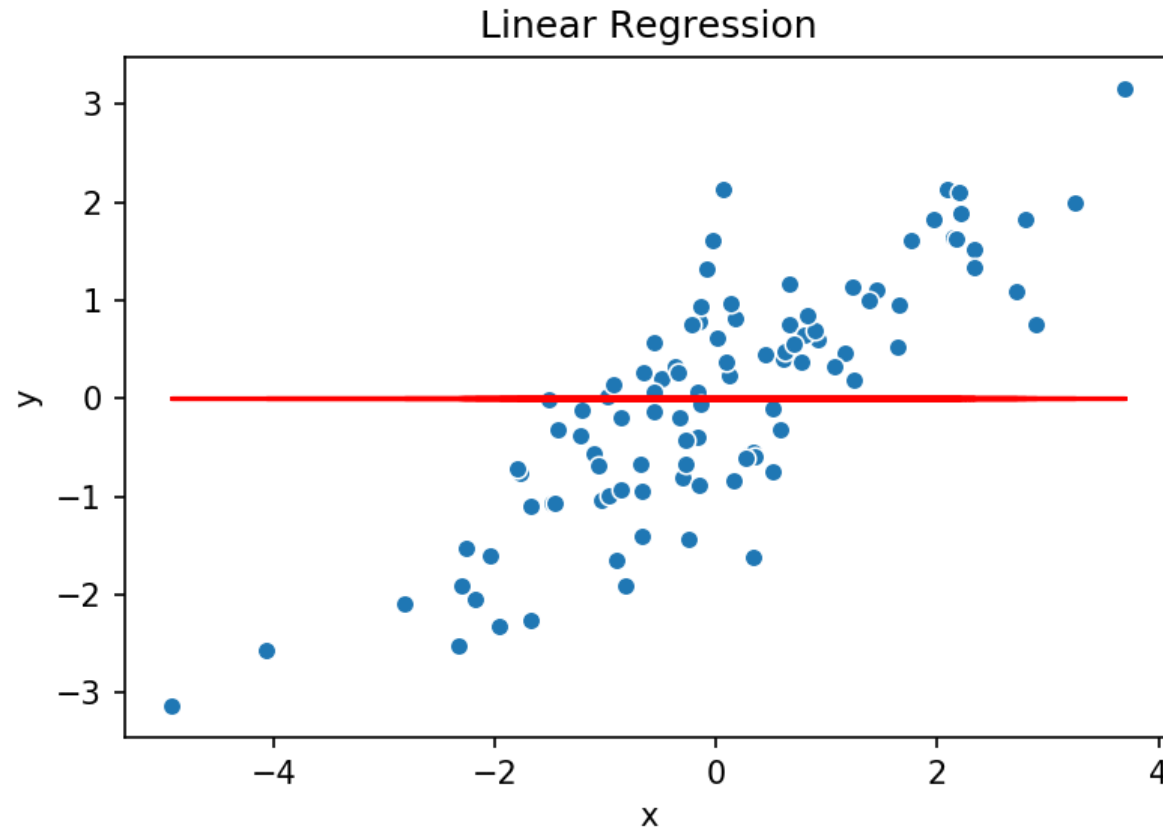
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Practice



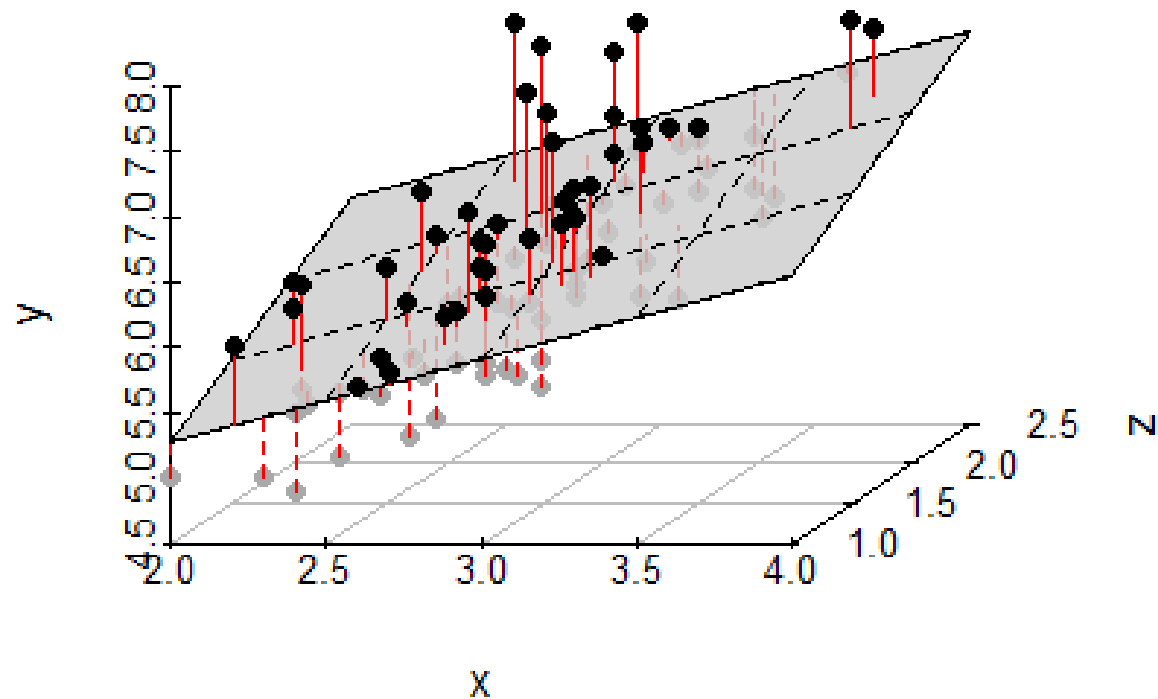
Linear Regression Example

- We minimize the sum of squared errors (SSE).



Linear Regression Example

- The vector of estimates can be fit with $\hat{\beta} = (X^T X)^{-1} (X^T y)$
- The predictions can be obtained with $\hat{y} = X\hat{\beta}$



Linear Regression Example

❑ Class implementation

```
class ISA591_LinearRegression:
    def __init__(self):
        self.X = X
        self.y = y

    def fit(self, X, y):
        Xt = np.transpose(X)
        XtX = np.dot(Xt, X) # X transpose X
        Xty = np.dot(Xt, y) # X tranpose y
        self.beta_hat = np.linalg.inv(XtX).dot(Xty)

        print(f'The y-intercept is {self.beta_hat[0]}
              and the slope is {self.beta_hat[1]}')

    def predict(self, X):
        return np.dot(X, self.beta_hat)
}
```

Linear Regression Example

❑ Class implementation

```
● ● ●  
  
## Make Instance  
my_reg = ISA590_LinearRegression()  
  
## Let's call the fit method with X and y  
my_reg.fit(X, y)  
  
## Let's call the predict method on X  
my_reg.predict(X)
```

Practice

